

Yuxuan Ni

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EDUCATION

The University of Alabama <i>Ph.D., Electrical and Computer Engineering Tuscaloosa, AL</i>	Sep 2025 - Present
Carnegie Mellon University <i>M.S. in Engineering & Technology Innovation Management Pittsburgh, PA</i>	Jan 2023 - Dec 2023
UESTC / University of Glasgow <i>B.E. in Communication Engineering; B.E. in Electronics and Electrical Engineering with Communications</i>	Sep 2018 - Jun 2022

SELECTED RESEARCH PROJECTS

Bunker Multi-Sensor SLAM, Localization, and Mapping <i>Research Project Prof. Shunqiao Sun and Prof. Igal Bilik The University of Alabama / Ben-Gurion University</i>	Aug 2025 - Present
- Built a ROS2-based UGV perception and evaluation stack for bunker, low-light, and GNSS-degraded field runs, synchronizing camera, LiDAR, 4D radar, INS/RTK, wheel feedback, and driver-control data. - Developed real-time LiDAR localization and soft-anchor mapping workflows; reduced a 459.5M-point raw scan stream to a 2.13M-point map (99.5% reduction) and cleaned a 24.29M-point map to 13.56M points. - Improved radar localization through IMU-aided fusion and sensor-specific parameter tuning, using INS/RTK alignment as the reference for trajectory evaluation and failure-case analysis. - Evaluated radar, visual, and LiDAR-inertial localization in a low-light, building-occluded sequence; radar odometry reached 7.16 m ATE RMSE versus 35.47 m for visual SLAM and 35.57 m for a LiDAR-inertial baseline. - Prepared camera-view, radar/LiDAR point-cloud, and top-view trajectory diagnostics to compare sensor behavior across bunker turns, occluded areas, and low-visibility segments.	

Learning-Based 4D Radar Odometry and Camera Fusion <i>Research Project Prof. Shunqiao Sun The University of Alabama</i>	Jan 2026 - Present
- Extended a learning-based 4D radar odometry pipeline for sparse radar point clouds, adding range/intensity pre-filtering and point-multiplication experiments to improve usable geometric structure before pose estimation. - Built two-stage odometry experiments that first stabilize radar pose outputs and then evaluate camera-pose fusion, including an MLP-based module for combining camera estimates with radar trajectories. - Added lightweight pseudo-correspondence supervision while preserving the original pose backbone; introduced 84.1K trainable parameters, about 0.53% of the full model, for correspondence and offset learning. - Reduced correspondence loss by 36.0%, match/static losses by about 36.8%, and offset error from 0.409 m to 0.363 m in frozen-backbone ablations; guarded refinement reduced translation RMSE from 0.752 m to 0.311 m. - Maintained frozen-backbone and guarded-refinement ablations to separate correspondence-learning gains from pose-backbone changes, keeping trajectory comparisons consistent across variants.	

NXP Radar DDM Decoding <i>NXP collaborative radar signal-processing project</i>	Aug 2025 - Present
- Developed a MATLAB SFW/DDM simulation and decoding pipeline for 12-TX/12-RX, 1024-sample, 512-chirp radar data with multi-TX de-spreading and range-Doppler fusion. - Implemented CA-CFAR and DBSCAN-based target extraction to separate DDM ghost artifacts, SFW Doppler offsets, and velocity-induced Doppler smearing in staged validation scenarios. - Designed velocity-search and quadratic phase-correction desmearing, validating a 320-candidate search on close targets with 2 m range separation and 40 m/s velocity difference.	

PRIOR RESEARCH EXPERIENCE

MVMD-Based CSP for EEG Signal Classification <i>Research Assistant, supervised by Prof. Aimin Jiang Hohai University</i>	Jun 2024 - Feb 2025
- Applied MVMD and CSP/ADMM-based filters across 1,000+ EEG datasets, improving classification accuracy by about 10% while reducing computational complexity by about 20%. - Compared MLP, LDA, and SVM classifiers to analyze inter- and intra-subject variability and validate feature robustness across EEG classification settings.	
NGSO Satellite In-line Interference Analysis <i>Research Assistant, supervised by Prof. Jon Peha and Prof. Marvin Sirbu Carnegie Mellon University</i>	Aug 2023 - Jan 2024
- Modeled satellite-ground communication links with TLE data in MATLAB and Python, processing million-scale datasets and reducing estimated interference probabilities by 30%. - Designed an interference-avoidance simulation workflow that reduced computational cost by about 60% while preserving scenario-level link and geometry analysis.	

WORK EXPERIENCE

CMU Center for Technology Transfer and Enterprise Creation <i>Enterprise Creation Intern Pittsburgh, PA</i>	May 2023 - Aug 2023
- Conducted customer discovery and market analysis for CMU Bio-robotic Lab and Phlux, connecting technical use cases with commercialization evidence. - Built customer profiles, screened high-potential leads, and synthesized interview findings to support venture exploration and technology-transfer decisions.	
Signify (China) Investment Co., Ltd. <i>Hardware Engineer Intern Shanghai, China</i>	Sep 2022 - Dec 2022
- Optimized power-circuit designs with active EMI filters and high-performance capacitors, reducing power consumption by 30%. - Debugged IoT lamp control logic and testing workflows, supporting hardware validation and embedded-device troubleshooting.	

TECHNICAL SKILLS

Python, MATLAB, C/C++, ROS/ROS2, Linux, PCL, OpenCV, PyTorch, NumPy.